

# MCE-based Direct FTC Method for Dynamic Positioning of Underwater Vehicles with Thruster Redundancy\*

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**Abstract**—This paper presents an active model-based FTC (fault-tolerant control) method for the dynamic positioning of a class of underwater vehicles with thruster redundancy. Compared to the widely used state and parameter estimation methods, this proposed scheme directly utilizes the vehicle's motion control error (MCE) to construct a residual for detecting thruster faults and failures in the steady state of the control system. In the case of thruster fault identification, the most difficult aspect is that the actual control input with thruster faults is unknown. However, through a detailed and precise analyses of MCE variation trends in the case of thruster faults, highly useful information about this unknown control input can be extracted. This characteristic also serves as the foundation for the novel scheme proposed in this paper. As for control reconfiguration, it is straightforward since the thrust losses can be directly estimated as a result of the identification process. Numerical studies with the real world vehicle model are also carried out to demonstrate the effectiveness of the proposed method.

## I. INTRODUCTION

Over the past few decades, interest in deep sea exploration has steadily grown, driven by its wide range of applications, including scientific research, resource exploration, oil and gas drilling, rescue operation, and various military activities. This insistent interest has fueled rapid advancements in underwater vehicle technology, further intensifying human curiosity about the deep sea.

Traditional ocean exploration typically consists of two consecutive steps of operations. The first one is a large-area survey conducted using torpedo-type autonomous underwater vehicles (AUVs), which are equipped with various sonar devices to inspect the seafloor environment. Once the AUV is recovered, recorded sonar images are manually analyzed to identify specific points of interest. The process then moves to the second step, where remotely operated vehicles (ROVs) or manned submersibles are deployed to these identified locations for detailed exploration and manipulation tasks, ultimately fulfilling the mission.

To ensure the reliability and safety of these expensive underwater robotic systems, ROVs and manned submersibles are typically designed with redundant thruster configurations. From a system science perspective, one of the most critical failure modes for these vehicles is thruster malfunction, alongside communication loss. As a result, the design of

fault-tolerant control (FTC) strategies to address thruster faults has emerged as a highly attractive research topic in robotics and control engineering.

This paper focuses on the model-based active FTC problem for a class of underwater vehicles with thruster redundancy. Compared to the related previous works, the main contributions of this work can be summarized as follows:

- The primary goal of thruster FTC is to ensure proper vehicle motion control. With this in mind, in this paper, motion control error (MCE) is directly applied to construct a residual for detecting fault and failure. This approach is significantly more compact compared to the previous works, which rely on estimation errors to generate residuals and therefore require an additional estimation scheme.
- A major challenge in thruster fault identification is the lack of knowledge about the actual control input when a fault occurs. However, through a detailed analysis of MCE variations under thruster faults, we demonstrate that valuable information about this unknown control input can be inferred—or even estimated. This insight plays a crucial role in the proposed fault identification scheme.
- Once thruster faults are identified through the proposed method, control reconfiguration or allocation becomes straightforward, as these corresponding thrust losses can be directly estimated as part of the identification process.

## II. RELATED WORKS

Fault detection and identification (FDI) methods based on analytical redundancy have been widely studied in various complex dynamics systems [1], [2]. For some of large scale systems where dynamic modeling is prohibitively time-consuming, model-free method, especially using the expert system and artificial neural network, is suitable for their FTC [2], [3]. However, this method usually requires a suitable training set data, which is difficult to be acquired for many real-world dynamics systems. In addition, training neural networks, whether in off-line [4] or on-line [5] learning cases, can be computationally intensive and time-consuming [2]. In contrast, model-based method leverages the idea of generating residuals that reflect the inconsistency between actual and estimated behavior [2], [6]. State-estimation and parameter-estimation methods are two of most common schemes in this group [7]–[9]. These methods apply various system identification techniques to estimate system states or model parameters and using corresponding estimation errors

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# 基于MCE的具有推进器冗余的水下航行器动态定位直接FTC方法\*

李吉弘

**Abstract**—— 本文提出了一种针对具有推进器冗余的水下车辆动态定位的主动基于模型的容错控制（FTC）方法。与广泛使用的状态和参数估计方法相比，该方案直接利用车辆的运动控制误差（MCE）来构建残差，以在控制系统稳态下检测推进器的故障和失效。在推进器故障识别的情况下，最难的方面是推进器故障下的实际控制输入未知。然而，通过对推进器故障情况下MCE变化趋势的详细而精确的分析，可以提取关于该未知控制输入的高度有用的信息。这一特性也作为本文提出的新方案的基础。至于控制重构，这相对直接，因为推进力损失可以通过识别过程直接估计。数值测量

## 一、引言

在过去的几十年里，人们对深海探索的兴趣稳步增长，这一兴趣的增长是由其广泛的应用所推动的，包括科学研究、资源勘探、石油和天然气钻探、救援行动以及各种军事活动。这种持续的兴趣推动了水下车辆技术的快速发展，进一步加深了人类对深海的好奇心。

传统的海洋勘探通常包括两个连续的操作步骤。第一步是使用鱼雷型自主水下航行器（AUV）进行大范围调查，这些航行器配备了各种声纳设备以检查海底环境。一旦AUV被回收，记录的声纳图像将被人工分析，以识别特定的兴趣点。随后，过程进入第二步，远程操作车辆（ROV）或有人潜水器会被部署到这些识别出的地点进行详细的勘探和操作任务，从而最终完成任务。

为了确保这些昂贵的水下机器人系统的可靠性和安全性，远程操作潜水器和载人潜水器通常设计有冗余推进器配置。从系统科学的角度来看，这些交通工具最关键的故障模式之一是推进器故障，以及通信中断。因此，其设计

为应对推进器故障而提出的容错控制（FTC）策略已成为机器人技术和控制工程中一个高度吸引人的研究课题。

本文聚焦于一类具有推进器冗余的水下车辆的基于模型的主动容错控制（FTC）问题。与相关的前人工作相比，本文的主要贡献可以总结如下：

- 推进器FTC的主要目标是确保车辆的运动控制得当。基于此，本文直接将运动控制误差（MCE）用于构建残差，以检测故障和失效。与之前依赖估计误差生成残差并因此需要额外估计方案的工作相比，该方法显著更为紧凑。
- 推进器故障识别的一个主要挑战是当故障发生时对实际控制输入的缺乏了解。然而，通过对推进器故障下MCE变化的详细分析，我们证明了可以推断甚至估计关于这一未知控制输入的有价值信息。这一见解在所提出的故障识别方案中起着至关重要的作用。
- 一旦通过所提出的方法识别出推进器故障，控制重构或分配就变得很直接，因为这些相应的推力损失可以作为识别过程的一部分直接估计。

## 二、相关工作

基于解析冗余的故障检测与识别（FDI）方法已在各种复杂动力系统中得到广泛研究[1], [2]。对于一些大型系统，由于动态建模耗时过长，无模型方法，尤其是使用专家系统和人工神经网络的方法，适合用于它们的容错控制（FTC）[2], [3]。然而，这种方法通常需要合适的训练数据集，而对于许多实际动力系统来说，这类数据难以获取。此外，无论是在离线[4]还是在线[5]学习情况下，训练神经网络都可能计算量大且耗时[2]。相比之下，基于模型的方法利用生成残差的思路，这些残差反映了实际行为与估计行为之间的不一致[2], [6]。状态估计和参数估计方法是该类别中最常见的两种方案[7]–[9]。这些方法应用各种系统辨识技术来估计系统状态或

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to construct proper residuals to detect and diagnosis the faults and failures.

This paper addresses the model-based active FTC problem for a class of underwater vehicles with thruster redundancy. Quite a number of related works have been done thus far [10]–[15]. In [11], state estimation error with all states measurable is used to detect the fault, and further an unknown input-observer is proposed to isolate this fault. In [12]–[14], a sliding-mode observer is used to estimate unmeasurable states and use the deviation of sliding surface of this observer in the steady state to detect and isolate the thruster fault and failure. For control allocation or reconfiguration, all the weight parameters for each thruster are adjusted iteratively until the residual minimization is ensured.

For most of practical underwater vehicles, especially expensive work-class ROVs and manned submersibles, a precise and reliable navigation system is the cornerstone of the entire vehicle system. These vehicles are usually equipped with various navigation systems (see [16] and references therein) combining with precious acoustic positioning system and navigation sensors, such as IMU (inertial measurement unit) and DVL (Doppler velocity log). Therefore, from a control engineering perspective, there is no need for additional observers to estimate the vehicle's states. On the other hand, in the case of redundant system, it's convenient to design a proper control method to guarantee the various performances in practice [17], [18]. Building on these two perspectives, in the authors' previous work [19], a novel scheme of thruster FTC for a class of underwater vehicles with thruster redundancy is proposed. Instead of relying on estimation errors, the scheme directly utilizes the vehicle's trajectory tracking error to construct a residual for detecting thruster faults and failures. For further fault identification, a novel diagnosis scheme is presented based on a detailed analysis of tracking error changing trends in response to variations in each thrust force in the steady state of control system. A key advantage of this approach is its ability to quickly and effectively pinpoint the faulty thruster by analyzing the trends of specific error combinations, such as 2D position and azimuth errors in the horizontal plane. As for control reconfiguration, the process is straightforward: the thrust loss parameter associated with the identified faulty thruster is adjusted until the residual falls below a specified threshold again.

This paper aims to extend the results from [19] to the case of dynamic positioning, with the following newly added contributions:

- Unlike [19], where no uncertainty term is considered in the vehicle's dynamics, this paper accounts for unknown bounded uncertainty terms, including significant sea currents, for its dynamic positioning. All these uncertainties are assumed to be unmatched.
- In [19], the residual after the thruster fault remains elevated until the thrust loss parameter is adjusted to the correct value. In contrast, after the initial spike at the moment of thruster fault occurrence, the residual in this paper automatically decreases below the specified fault

detection criterion. For this reason, the method in [19] is not directly applicable to the case of dynamic positioning considered in this paper. However, this automatic decrease in residual also reveals an important property: the actual control input after thruster fault remains the same or very similar to the control input before the fault. Since this input matches the designed control input, it is known to us. Using this estimated actual control input after thruster fault, we can easily formulate the identification method and thrust loss estimation process. And this represents a key contribution of this paper.

- Furthermore, with this formulated identification method, the restrictive condition of *Assumption 3* in [19], which states that "at each time, only one thruster occurs fault or failure," is removed in this paper. The proposed method in this paper can identify up to two simultaneous double thruster faults and can accurately estimate the thrust losses of each of them.

### III. METHODOLOGY

#### A. Problem Statement

1) *Vehicle Model*: For most underwater vehicles, especially for work-class ROVs and manned submersibles, the center of gravity is often designed to be (much) lower than the center of buoyancy to stabilize roll motion. In addition, the thruster system is typically configured to allow easy decoupling of the vehicle's horizontal and vertical motions, although vertical thrusters are sometimes used for tilt control. Considering these aspects and for the sake of discussion, this paper focuses solely on the vehicle's 3DOF horizontal motion, with its kinematics and dynamics presented as follows [20].

$$\dot{\eta} = J\nu, \quad (1)$$

$$M\dot{\nu} + C\nu + D\nu + g(\eta) = \tau + d, \quad (2)$$

where  $\eta = [x, y, \psi]^T$  denotes the vehicle configuration on the horizontal plane in the navigation frame, and  $\nu = [u, v, r]^T$  is the velocity vector in the vehicle's body-fixed frame, see Fig. 1;  $M$  indicates the mass term including added mass,  $C$  denotes the Coriolis and centripetal term,  $D$  is the damping components, and  $g$  presents gravitational term, all of which satisfy to be of class  $C^2$ ;  $\tau = [\tau_u, \tau_v, \tau_r]^T$  is the thrust force and moment vector;  $d$  denotes the uncertainty term including modeling errors, measurement noises, exogenous disturbances such as sea currents, etc., all of which are unmatched;  $J$  denotes the coordinate transformation matrix from the vehicle's body-fixed frame to the navigation frame, and can be expresses as follows,

$$J = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

For uncertainty term  $d$ , we have the following assumption.

*Assumption 1.*  $|d_i| \leq D_{mi}$ ,  $i = 1, 2, 3$  with  $D_{mi} > 0$  unknown constant.

Considering (2), the vehicle's dynamics usually satisfies the following properties [20], [21]

构建适当的残差以检测和诊断故障和失效。

本文针对具有推进器冗余的一类水下航行器的基于模型的主动容错控制（FTC）问题进行了研究。迄今为止，已经完成了相当多的相关工作 [10]–[15]。在文献 [11] 中，使用所有状态可测时的状态估计误差来检测故障，并进一步提出了一个未知输入观测器以隔离该故障。在文献 [12]–[14] 中，使用滑模观测器来估计不可测量的状态，并利用该观测器在稳态时滑动表面的偏差来检测和隔离推进器故障和失效。对于控制分配或重新配置，每个推进器的所有权重参数都被迭代调整，直到确保残差最小化为止。

对于大多数实用水下车辆，尤其是昂贵的工作级ROV和载人潜水器，一个精确且可靠的导航系统是整个车辆系统的基石。这些车辆通常配备各种导航系统（参见[16]及其中引用的文献），结合精密的声学定位系统和导航传感器，如IMU（惯性测量单元）和DVL（多普勒速度计）。因此，从控制工程的角度来看，没有必要额外增加观测器来估计车辆状态。另一方面，在冗余系统的情况下，设计适当的控制方法以保证各种性能在实际中实现是方便的[17], [18]。基于这两个角度，在作者之前的工作[19]中，提出了一种针对具有推进器冗余的一类水下车辆的新型推进器故障容错控制（FTC）方案。该方案不依赖于估计误差，而是直接利用车辆的

检测标准。出于这个原因，文献[19]中的方法不能直接应用于本文所考虑的动态定位情况。然而，这种残差的自动减小也揭示了一个重要特性：推进器故障后的实际控制输入保持不变或与故障前的控制输入非常相似。由于该输入与设计的控制输入相匹配，所以我们是已知的。利用该估计的推进器故障后的实际控制输入，我们可以轻松地制定识别方法和推力损失估计过程。这也代表了本文的一个关键贡献。

- 此外，使用这种制定的识别方法，本论文中移除了文献[19]中的约束条件*Assumption 3*，该条件规定“在每个时间点，只有一个推进器发生故障或失效”。本论文提出的方法可以识别最多两台推进器同时发生的双重故障，并能够准确估计每台推进器的推力损失。

### 三、方法论

#### A. Problem Statement

1) *Vehicle Model*: 对于大多数水下航行器，尤其是工作级ROV和载人潜水器，重心通常被设计得比浮力中心低（很多），以稳定横滚运动。此外，推进器系统通常配置为允许车辆的水平运动和垂直运动轻松解耦，尽管有时垂直推进器用于倾斜控制。考虑到这些方面，并为了讨论的方便，本文仅关注车辆的三自由度水平运动，其运动学和动力学如下所示[20]。

$$\dot{\eta} = J\nu, \quad (1)$$

$$M\dot{\nu} + C\nu + D\nu + g(\eta) = \tau + d, \quad (2)$$

其中  $\eta = [x, y, \psi]^T$  表示导航坐标系下车辆在水平面上的构型， $\nu = [u, v, r]^T$  表示车辆机体系下的速度矢量，见图 1； $M$  表示包含附加质量的质量项， $C$  表示科里奥利力和向心力项， $D$  表示阻尼分量， $g$  表示重力项，所有这些都是  $C^2$  类； $\tau = [\tau_u, \tau_v, \tau_r]^T$  是推力力和力矩矢量； $d$  表示包括建模误差、测量噪声、外部扰动如海流等在内的不确定性项，这些都是未匹配的； $J$  表示从车辆机体系到导航坐标系的坐标变换矩阵，其表达式如下，

$$J = \begin{bmatrix} \cos \psi & -\sin \psi & 0 \\ \sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

对于不确定性项  $d$ ，我们有以下假设。 *Assumption 1*.  $|d_i| \leq D_{mi}$ ,  $i = 1, 2, 3$ ,  $D_{mi} > 0$  为未知常数。

考虑到 (2)，车辆的动力学通常满足以下特性 [20], [21]

本文旨在将[19]的结果扩展到动态定位的情形，并具有以下新增加的贡献：

- 与[19]不同，后者在车辆动力学中未考虑不确定性项，本文考虑了未知的有界不确定性项，包括对其动态定位有重要影响的海流。所有这些不确定性都被假定为不匹配。
- 在[19]中，推进器故障后的残差保持在较高水平，直到推力损失参数调整到正确值。相比之下，在推进器故障发生的瞬间出现初始峰值后，本文中的残差会自动下降到规定的故障值以下。

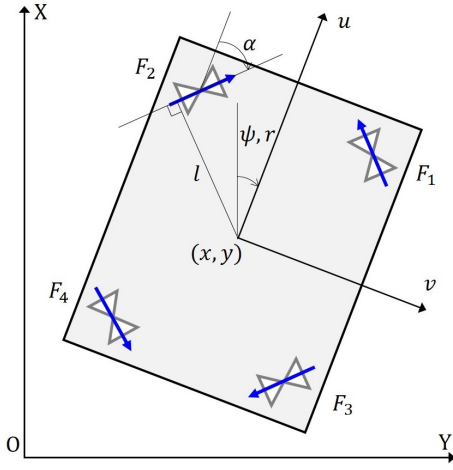


Fig. 1. Vehicle configuration in the navigation frame and its thruster configuration in the body-fixed frame.

**P1.**  $M = M^T > 0$ , and  $\dot{M}(t) = 0$ ,  $\forall t > 0$ .

**P2.**  $s^T [\dot{M} - 2C] s = 0$ ,  $\forall s \in \mathbb{R}^3$ .

**P3.**  $D > 0$ .

Unlike previous related works [12]–[14], where the control laws are derived in the navigation frame, this paper directly solves the control problem in the vehicle's body-fixed frame as in (2).

For thruster configuration, we consider the case as shown in Fig. 1. This is also the most common configuration for underwater vehicles, where the arrangement of four horizontal thrusters has a symmetrical structure in the fore-aft and left-right directions. In this case, the control input  $\tau = [\tau_u, \tau_v, \tau_r]^T$  can be presented as follows:

$$\tau = T_{conf} F$$

$$= \begin{bmatrix} \cos \alpha & \cos \alpha & -\cos \alpha & -\cos \alpha \\ -\sin \alpha & \sin \alpha & -\sin \alpha & \sin \alpha \\ -l & l & l & -l \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}, \quad (3)$$

where  $\alpha$  is the thruster orientation in the body-fixed frame and  $l$  denotes the distance from the center point to the thruster's orientation line.

**2) Thruster Fault and Failure Model:** For each thruster, its thrust output depends on the specific thruster model,

$$F_i = F_T(u_i), \quad (4)$$

where  $u_i$  is the thruster actual control input. Indeed, this is the input signal set into the thruster control board. In the case of BlueROV2 Heavy ROV [22], whose model will be used in the numerical studies in this paper, the thruster model (4) can be embodied as

$$F_i = K_i u_i, \quad (5)$$

with  $K_i$  the thrust coefficient. Here the sign of  $u_i$  indicates the thrust force direction: If  $u_i > 0$ , the thruster outputs forward thrust; otherwise, reverse thrust.

For thruster fault and failure model, this paper directly

implements the same model used in [10], [11] as follows:

$$F_i = K_i W_i u_i, \quad (6)$$

where  $W_i$  is a numerical weight indicating the thrust loss and defined as follows,

$$W_i = \begin{cases} 0 & \text{if } i\text{-th thruster has a failure} \\ 0 < W_i < 1 & \text{if } i\text{-th thruster is faulty} \\ 1 & \text{if } i\text{-th thruster is normal} \end{cases} \quad (7)$$

Consequently, the thruster force vector  $F$  can be expressed as follows:

$$F = KWu, \quad (8)$$

where  $K = \text{diag}(K_1, \dots, K_4)$ ,  $W = \text{diag}(W_1, \dots, W_4)$ , and  $u = [u_1, \dots, u_4]^T$ .

Suppose  $\tau_c$  is the control input calculated by the controller, then control allocation [23] is to allocate the actual control input  $u$  as follows:

$$u = W^{-1} K^{-1} T_{conf}^\dagger \tau_c, \quad (9)$$

where superscript “ $\dagger$ ” denotes the pseudo-inverse of a non-square matrix.

Here the problem is that, in practice, the exact value of  $W$  is unknown, and therefore the control allocation has to take the following form

$$\hat{u} = \hat{W}^{-1} K^{-1} T_{conf}^\dagger \tau_c, \quad (10)$$

where  $\hat{W}$  is the estimation of  $W$  and  $\hat{u}$  is the corresponding control allocation.

Consequently, the embodied control input  $\tau$  becomes

$$\tau = T_{conf} KW \hat{W}^{-1} K^{-1} T_{conf}^\dagger \tau_c. \quad (11)$$

If  $\hat{W} = W$ , then we have  $\tau = \tau_c$ . Otherwise, the deviation of  $\hat{W}$  from  $W$  can cause nonzero  $\Delta\tau = \tau_c - \tau$ , which further result in a similar deviation in the MCE during the steady state of the control system.

**3) Problem Formulation:** As in [19], this paper aims to answer the question, “Is it possible to detect the change in  $W$  and further identify which  $W_i$  is changed, by solely observing the deviation of MCE during the steady state of the control system?”

To address this problem, this paper supposes the following constraint conditions.

**Assumption 2.** Thruster fault and failure occur only after the control system has been stabilized.

**Assumption 3.** Thruster fault and failure is caused only by the reduction of corresponding numerical scale weight  $W$ .

Compared to [19], we can see that Assumption 3 in [19], which might be the most restricting condition in practical applications, is eliminated in this paper.

## B. Motion Control Design

Let  $\eta_d(t)$  denotes the vehicle's reference configuration and corresponding MCE is defined as  $e_\eta(t) = \|\eta_d(t) - \eta(t)\|_2$ . Here, if  $\dot{\eta}_d = 0$ , then this is regulation control problem, otherwise, becomes trajectory tracking problem. The control objective in this paper is to design a control law for  $\tau$  so that

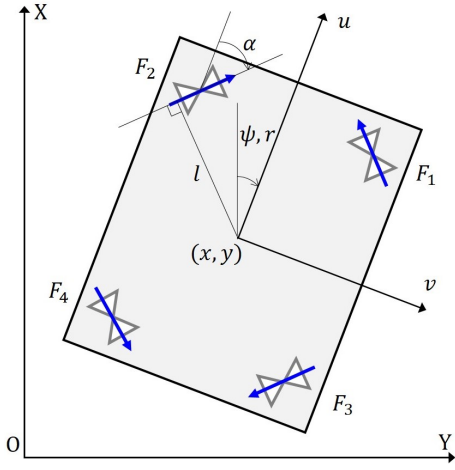


图 1. 导航参考系中的车辆配置及车体固定参考系中的推进器配置。

P1.  $M = M^T > 0$ , and  $\dot{M}(t) = 0$ ,  $\forall t > 0$ .

P2.  $s^T [\dot{M} - 2C] s = 0$ ,  $\forall s \in \mathbb{R}^3$ .

P3.  $D > 0$ .

与之前相关的工作 [12]–[14] 不同，那些工作是在导航坐标系下推导控制律的，本文直接在车辆本体固定坐标系中如公式 (2) 所示解决控制问题。

对于推进器配置，我们考虑如图1所示的情况。这也是水下车辆最常见的配置，其中四个水平推进器的布置在前后和左右方向上具有对称结构。在这种情况下，控制输入  $\tau = [\tau_u, \tau_v, \tau_r]^T$  可以表示如下：

$$\tau = T_{conf} F$$

$$= \begin{bmatrix} \cos \alpha & \cos \alpha & -\cos \alpha & -\cos \alpha \\ -\sin \alpha & \sin \alpha & -\sin \alpha & \sin \alpha \\ -l & l & l & -l \end{bmatrix} \begin{bmatrix} F_1 \\ F_2 \\ F_3 \\ F_4 \end{bmatrix}, \quad (3)$$

其中  $\alpha$  是机体固定坐标系下的推进器方向， $l$  表示从中心点到推进器方向线的距离。

2) *Thruster Fault and Failure Model*: 对于每个推进器，其推力输出取决于特定的推进器型号，

$$F_i = F_T(u_i), \quad (4)$$

其中  $u_i$  是推进器的实际控制输入。实际上，这是设置到推进器控制板的输入信号。在本文数值研究中将使用其模型的 BlueROV2 Heavy ROV [22] 的情况下，推进器模型 (4) 可以表示为

$$F_i = K_i u_i, \quad (5)$$

与  $K_i$  一起为推力系数。这里  $u_i$  的符号表示推力方向：如果  $u_i > 0$ ，推进器输出正向推力；否则，输出反向推力。

对于推进器故障和失效模型，本文直接

实现了与[10]、[11]中使用的相同模型，如下所示：

$$F_i = K_i W_i u_i, \quad (6)$$

其中  $W_i$  是表示推力损失的数值权重，定义如下，

$$W_i = \begin{cases} 0 & \text{if } i\text{-th thruster has a failure} \\ 0 < W_i < 1 & \text{if } i\text{-th thruster is faulty} \\ 1 & \text{if } i\text{-th thruster is normal} \end{cases} \quad (7)$$

因此，推进器力矢量  $F$  可表示如下：

$$F = KWu, \quad (8)$$

哪里  $K = \text{diag}(K_1, \dots, K_4)$ 、 $W = \text{diag}(W_1, \dots, W_4)$  和  $u = [u_1, \dots, u_4]^T$ 。

假设  $\tau_c$  是控制器计算的控制输入，那么控制分配 [23] 是将实际控制输入  $u$  分配如下：

$$u = W^{-1} K^{-1} T_{conf}^\dagger \tau_c, \quad (9)$$

上标 “ $\dagger$ ” 表示非方阵的伪逆。

这里的问题是，在实际情况下， $W$  的确切值是未知的，因此控制分配必须采取以下形式

$$\hat{u} = \hat{W}^{-1} K^{-1} T_{conf}^\dagger \tau_c, \quad (10)$$

其中  $\hat{W}$  是对  $W$  的估计， $\hat{u}$  是相应的控制分配。

因此，具体的控制输入  $\tau$  变成

$$\tau = T_{conf} KW \hat{W}^{-1} K^{-1} T_{conf}^\dagger \tau_c. \quad (11)$$

如果  $\hat{W} = W$ ，那么我们有  $\tau = \tau_c$ 。否则， $\hat{W}$  与  $W$  的偏差可能导致非零的  $\Delta\tau = \tau_c - \tau$ ，这进一步会在控制系统的稳态下引起 MCE 的类似偏差。

3) *Problem Formulation*: 如同在[19]中，这篇论文旨在回答问题，“Is it possible to detect the change in  $W$  and further identify which  $W_i$  is changed, by solely observing the deviation of MCE during the steady state of the control system?”

为了解决这个问题，本文假设以下约束条件。

*Assumption 2* 推进器故障和失效仅在控制系统稳定后才会发生。

*Assumption 3* 推进器故障和失效仅由相应数值刻度权重  $W$  的降低引起。

与[19]相比，我们可以看到，本论文中消除了[19]中的假设3，而假设3可能是实际应用中最具限制性的条件。

## B. Motion Control Design

设  $\eta_d(t)$  表示车辆的参考配置，并且对应的 MCE 定义为  $e_\eta(t) = \|\eta_d(t) - \eta(t)\|_2$ 。在这里，如果  $\dot{\eta}_d = 0$ ，则这是调控控制问题，否则，成为轨迹跟踪问题。本文的控制目标是为  $\tau$  设计控制律，以便

$e_\eta(t)$  to be uniformly ultimately bounded (UUB) with  $t \rightarrow \infty$ . For this purpose, the vehicle's kinematics and dynamics can be rewritten as follows:

$$\dot{e}_\eta = \eta_d - J\nu, \quad (12)$$

$$M\dot{\nu} = -C\nu - D\nu - g(\eta) + \tau + d. \quad (13)$$

For this second-order strict-feedback form of motion control system, general backstepping method [24] is utilized to solve the control problem.

1) *Step 1:* For given MCE  $e_\eta(t)$ , the following Lyapunov function candidate is considered in this step,

$$V_1 = \frac{1}{2} e_\eta^T \Gamma_1 e_\eta, \quad (14)$$

where  $\Gamma_1 > 0$  is a diagonal weighting matrix.

By differentiating (14) and substituting (13) into it, we get

$$\dot{V}_1 = e_\eta^T \Gamma_1 (\dot{\eta}_d - J\nu), \quad (15)$$

according to which, the following control law is chosen for  $\alpha_\nu$ , which is the stabilization function [24] for virtual control input  $\nu$  in (15),

$$\alpha_\nu = J^{-1} (\dot{\eta}_d + \Gamma_1^{-1} A_1 e_\eta), \quad (16)$$

where  $A_1 > 0$  is diagonal control gain matrix.

By substituting (16) into (15), we have

$$\dot{V}_1 = -e_\eta^T A_1 e_\eta + e_\eta^T \Gamma_1 J e_\nu, \quad (17)$$

where  $e_\nu = \alpha_\nu - \nu = [e_u, e_v, e_r]^T$ .

2) *Step 2:* Considering the vehicle's dynamics as (13) with properties P1–P3, the following Lyapunov function candidate is taken in this step.

$$V_2 = V_1 + \frac{1}{2} [e_\nu^T M e_\nu + \tilde{D}_m^T \Gamma_2 \tilde{D}_m], \quad (18)$$

where  $\tilde{D}_m = \hat{D}_m - D_m$  with  $D_m = [D_{m1}, D_{m2}, D_{m3}]^T$  and  $\hat{D}_m$  the estimation of  $D_m$ , and  $\Gamma_2 > 0$  is a design diagonal weighting matrix.

By differentiating (18) and substituting (17) and (13) into it, we have

$$\begin{aligned} \dot{V}_2 = & -e_\eta^T A_1 e_\eta + e_\eta^T \Gamma_1 J e_\nu + \frac{1}{2} e_\nu^T \dot{M} e_\nu \\ & + e_\nu^T [M \dot{\alpha}_\nu + (C + D)\nu + g(\nu) - \tau - d] \\ & + \tilde{D}_m^T \Gamma_2 \dot{\tilde{D}}_m. \end{aligned} \quad (19)$$

According to (19), we choose the control law for  $\tau$  and corresponding adaptation law for  $\hat{D}_m$  as follows:

$$\tau = A_2 e_\nu + \Gamma_1 J e_\eta + M \dot{\alpha}_\nu + (C + D)\alpha_\nu + g(\eta) + \hat{d}, \quad (20)$$

$$\dot{\hat{D}}_m = \Gamma_2^{-1} e_\nu, \quad (21)$$

where  $A_2 > 0$  is diagonal control gain matrix, and  $\hat{d} \in \mathbb{R}^3$  with  $\hat{d}_i = \hat{D}_{mi} \varphi(\hat{D}_{mi} e_{\nu(i)})$ . Here, the smooth function  $\varphi(\cdot)$  satisfies the following lemma [25].

*Lemma 1.*  $\forall \epsilon > 0$ , there exist a smooth function  $\varphi(\cdot)$  such that  $\varphi(0) = 0$  and

$$|\xi| \leq \xi \varphi(\xi) + \epsilon, \quad \forall \xi \in \mathbb{R}. \quad (22)$$

*Remark 1.* One example of smooth function satisfying Lemma 1 is  $\varphi(\xi) = \tanh(\kappa \xi / \epsilon)$  with  $\kappa$  the solution of the equation  $\kappa = e^{-(\kappa+1)}$  [25].

*Theorem 1.* Consider the trajectory tracking problem for (1) and (2). If we choose the control law as (20) and corresponding adaptation law as (21), then we can guarantee the closed-loop control system to be UUB.

*Proof.* By substituting (20) and (21) into (19), we can get

$$\begin{aligned} \dot{V}_2 = & -e_\eta^T A_1 e_\eta - e_\nu^T A_2 e_\nu + \frac{1}{2} (\dot{M} - 2C) e_\nu \\ & - e_\nu^T D e_\nu - e_\nu^T d - e_\nu^T \hat{d} + \tilde{D}_m^T e_\nu \\ \leq & -e_\eta^T A_1 e_\eta - e_\nu^T A_2 e_\nu + \epsilon_1 + \epsilon_2 + \epsilon_3. \end{aligned} \quad (23)$$

In the final expansion, Lemma 1 is applied with  $\epsilon_1, \epsilon_2, \epsilon_3 > 0$  the design parameters defined as in Lemma 1. (23) indicates that both of  $e_\eta$  and  $e_\nu$  are all bounded by the constants  $\epsilon_1, \epsilon_2, \epsilon_3 > 0$ , and this concludes the proof.  $\square$

*Remark 2.* Since  $\epsilon_1, \epsilon_2, \epsilon_3 > 0$  can be chosen arbitrarily small, it's possible for us to get arbitrarily small value of MCE  $e_\eta$  and  $e_\nu$ . However, too small value of  $\epsilon_1, \epsilon_2$ , and  $\epsilon_3$  may cause certain high-gain control problem. Therefore, these parameters should be carefully chosen in practice.

### C. Fault Detection and Identification

In practical applications, the thrusters are initially presumed to be functioning normally. In this case, we have  $\hat{W} = W = \text{diag}(1, \dots, 1)$ . Suppose there is a thrust loss  $\Delta W$  at  $t = t_c > 0$ . In this case,  $W = \hat{W} - \Delta W$ , combining with which, (11) can be expanded as follows:

$$\begin{aligned} \tau &= T_{conf} K W \hat{W}^{-1} K^{-1} T_{conf}^+ \tau_c \\ &= \overbrace{\begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 \\ -\tau_1 & \tau_2 & \tau_3 & -\tau_4 \\ -a\tau_1 & a\tau_2 & -a\tau_3 & a\tau_4 \end{bmatrix}}^{M_\tau} \begin{bmatrix} 1 - \Delta W_1 / \hat{W}_1 \\ \vdots \\ 1 - \Delta W_4 / \hat{W}_4 \end{bmatrix} \\ &= \tau_c - M_\tau \delta W, \end{aligned} \quad (24)$$

where  $\delta W \in \mathbb{R}^4$  with  $\delta W(i) = \Delta W_i / \hat{W}_i$ ,  $i = 1, \dots, 4$ , and

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1/a \\ 1 & 1 & 1/a \\ 1 & 1 & -1/a \\ 1 & -1 & 1/a \end{bmatrix} \tau_c = M_c^T \tau_c, \quad (25)$$

with  $a = l \sec \alpha = 0.267$ . In the above expansion, for the sake of simplicity, we directly apply the parameters  $\alpha = \pi/4$  and  $l = 0.1888m$  of BlueROV2 Heavy ROV model [22].

*Proposition 1.* If the designed controller  $\tau_c$  calculated as (20) has no component equal to zero, then  $M_\tau$  has rank 3.

*Proof.* Since  $M_c$  has rank 3,  $\tau_c$  with non component equal to zero can guarantee at least three of  $\tau_i$ ,  $i = 1, \dots, 4$  are nonzero, which further guarantees  $M_\tau$  to be rank 3.  $\square$

*Remark 3.* Here, it's worth to mention that all components of  $\tau_c$  are nonzero is a sort of sufficient condition, not necessary one. It's easy to see that any of  $\tau_c \in \{[\zeta, 0, 0]^T, [0, \zeta, 0]^T, [0, 0, \zeta]^T\}$  with  $\zeta \neq 0$  can guarantee all

$e_\eta(t)$  关于具有  $t \rightarrow \infty$  的一致最终有界 (UUB)。为此, 车辆的运动学和动力学可以重写如下:

$$\dot{e}_\eta = \eta_d - J\nu, \quad (12)$$

$$M\dot{\nu} = -C\nu - D\nu - g(\eta) + \tau + d. \quad (13)$$

对于这种二阶严格反馈形式的运动控制系统, 采用通用反步法[24]来解决控制问题。

1) *Step 1*: 对于给定的 MCE  $e_\eta(t)$ , 在此步骤中考虑以下 Lyapunov 函数候选。

$$V_1 = \frac{1}{2} e_\eta^T \Gamma_1 e_\eta, \quad (14)$$

其中  $\Gamma_1 > 0$  是一个对角加权矩阵。

通过对(14)求导并将(13)代入其中, 我们得到

$$\dot{V}_1 = e_\eta^T \Gamma_1 (\dot{\eta}_d - J\nu), \quad (15)$$

根据该方法, 为  $\alpha_\nu$  选择了以下控制律, 它是 (15) 中虚拟控制输入  $\nu$  的稳定化函数 [24],

$$\alpha_\nu = J^{-1} (\dot{\eta}_d + \Gamma_1^{-1} A_1 e_\eta), \quad (16)$$

其中  $A_1 > 0$  是对角控制增益矩阵。

通过将 (16) 代入 (15), 我们得到

$$\dot{V}_1 = -e_\eta^T A_1 e_\eta + e_\eta^T \Gamma_1 J e_\nu, \quad (17)$$

在  $e_\nu = \alpha_\nu - \nu = [e_u, e_v, e_r]^T$ 。

2) *Step 2*: 考虑车辆的动力学如式(13)所示, 具有属性P1-P3, 在这一步中选取以下Lyapunov函数候选。

$$V_2 = V_1 + \frac{1}{2} [e_\nu^T M e_\nu + \tilde{D}_m^T \Gamma_2 \tilde{D}_m], \quad (18)$$

其中  $\tilde{D}_m = \hat{D}_m - D_m$  与  $D_m = [D_{m1}, D_{m2}, D_{m3}]^T$  和  $\hat{D}_m$  对  $D_m$  的估计, 以及  $\Gamma_2 > 0$  是一个设计对角加权矩阵。

通过对(18)求导并将(17)和(13)代入其中, 我们得到

$$\begin{aligned} \dot{V}_2 = & -e_\eta^T A_1 e_\eta + e_\eta^T \Gamma_1 J e_\nu + \frac{1}{2} e_\nu^T \dot{M} e_\nu \\ & + e_\nu^T [M \dot{\alpha}_\nu + (C + D)\nu + g(\nu) - \tau - d] \\ & + \tilde{D}_m^T \Gamma_2 \dot{\tilde{D}}_m. \end{aligned} \quad (19)$$

根据 (19), 我们选择  $\tau$  的控制律和  $\hat{D}_m$  的相应自适应律如下:

$$\tau = A_2 e_\nu + \Gamma_1 J e_\eta + M \dot{\alpha}_\nu + (C + D)\alpha_\nu + g(\eta) + \hat{d}, \quad (20)$$

$$\dot{\hat{D}}_m = \Gamma_2^{-1} e_\nu, \quad (21)$$

其中  $A_2 > 0$  是对角控制增益矩阵,  $\hat{d} \in \mathbb{R}^3$  与  $\hat{d}_i = \hat{D}_{mi} \varphi(\hat{D}_{mi} e_{\nu(i)})$ 。这里, 平滑函数  $\varphi(\cdot)$  满足以下引理 [25]。

*Lemma 1.*  $\forall \epsilon > 0$ , 存在一个光滑函数  $\varphi(\cdot)$  使得  $\varphi(0) = 0$  并且

$$|\xi| \leq \xi \varphi(\xi) + \epsilon, \quad \forall \xi \in \mathbb{R}. \quad (22)$$

*Remark 1.* 满足引理1的光滑函数的一个例子是  $\varphi(\xi) = \tanh(\kappa \xi / \epsilon)$ , 其中  $\kappa$  是方程  $\kappa = e^{-(\kappa+1)}$  的解 [25]。

*Theorem 1* 考虑 (1) 和 (2) 的轨迹跟踪问题。如果我们选择控制律为 (20) 并且相应的自适应律为 (21), 那么我们可以保证闭环控制系统是 UUB 的。

*Proof* 通过将(20)和(21)代入(19), 我们可以得到

$$\begin{aligned} \dot{V}_2 = & -e_\eta^T A_1 e_\eta - e_\nu^T A_2 e_\nu + \frac{1}{2} (\dot{M} - 2C) e_\nu \\ & - e_\nu^T D e_\nu - e_\nu^T d - e_\nu^T \hat{d} + \tilde{D}_m^T e_\nu \\ \leq & -e_\eta^T A_1 e_\eta - e_\nu^T A_2 e_\nu + \epsilon_1 + \epsilon_2 + \epsilon_3. \end{aligned} \quad (23)$$

在最终的展开式中, 命题 1 应用于  $\epsilon_1, \epsilon_2, \epsilon_3 > 0$ , 设计参数如命题 1 中定义。(23) 表明  $e_\eta$  和  $e_\nu$  都被常数  $\epsilon_1, \epsilon_2, \epsilon_3 > 0$  所界定, 这就完成了证明。□

*Remark 2.* 由于  $\epsilon_1, \epsilon_2, \epsilon_3 > 0$  可以任意小地选择, 我们可以得到 MCE  $e_\eta$  和  $e_\nu$  的任意小值。然而,  $\epsilon_1, \epsilon_2$  和  $\epsilon_3$  的值过小可能会引发某些高增益控制问题。因此, 在实际中应谨慎选择这些参数。

### C. Fault Detection and Identification

在实际应用中, 推动器最初假定正常工作。在这种情况下, 我们有  $\hat{W} = W = \text{diag}(1, \dots, 1)$ 。假设在  $t = t_c > 0$  处发生推力损失  $\Delta W$ 。在这种情况下,  $W = \hat{W} - \Delta W$ , 结合上述内容, (11) 可以展开如下:

$$\begin{aligned} \tau &= T_{conf} K W \hat{W}^{-1} K^{-1} T_{conf}^\dagger \tau_c \\ &= \overbrace{\begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 \\ -\tau_1 & \tau_2 & \tau_3 & -\tau_4 \\ -a\tau_1 & a\tau_2 & -a\tau_3 & a\tau_4 \end{bmatrix}}^{M_\tau} \begin{bmatrix} 1 - \Delta W_1 / \hat{W}_1 \\ \vdots \\ 1 - \Delta W_4 / \hat{W}_4 \end{bmatrix} \\ &= \tau_c - M_\tau \delta W, \end{aligned} \quad (24)$$

在  $\delta W \in \mathbb{R}^4$  与  $\delta W(i) = \Delta W_i / \hat{W}_i$ ,  $i = 1, \dots, 4$ , 以及

$$\begin{bmatrix} \tau_1 \\ \tau_2 \\ \tau_3 \\ \tau_4 \end{bmatrix} = \begin{bmatrix} 1 & -1 & -1/a \\ 1 & 1 & 1/a \\ 1 & 1 & -1/a \\ 1 & -1 & 1/a \end{bmatrix} \tau_c = M_c^T \tau_c, \quad (25)$$

使用  $a = l$  秒  $\alpha = 0.267$ 。在上述展开中, 为了简单起见, 我们直接应用 BlueROV2 Heavy ROV 模型 [22] 的参数  $\alpha = \pi/4$  和  $l = 0.1888m$ 。

*Proposition 1* 如果设计的控制器  $\tau_c$  按 (20) 计算时没有任何组件等于零, 那么  $M_\tau$  的秩为 3。

*Proof.* 由于  $M_c$  的秩为 3,  $\tau_c$  的非零分量不为零可以保证  $\tau_i, i = 1, \dots, 4$  中至少三个不为零, 这进一步保证了  $M_\tau$  的秩为 3。□

*Remark 3* 在这里, 值得一提的是,  $\tau_c$  的所有分量都是非零的这一点是一种充分条件, 而不是必要条件。很容易看出, 任何  $\tau_c \in \{[\zeta, 0, 0]^T, [0, \zeta, 0]^T, [0, 0, \zeta]^T\}$  与  $\zeta \neq 0$  都可以保证所有

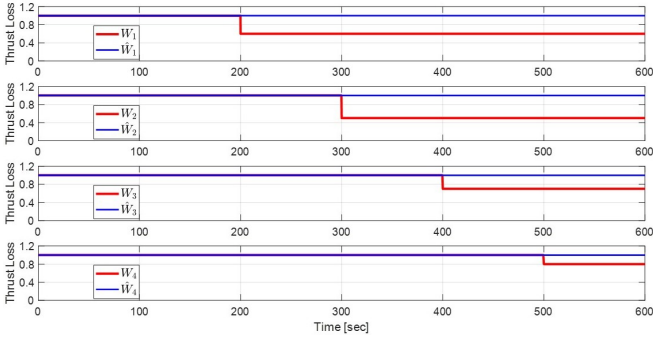


Fig. 2. Thruster fault occurrence scenario.

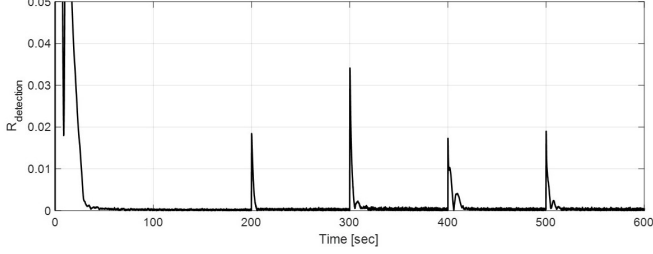


Fig. 3. Corresponding residual measurement.

$\tau_i$  to be nonzero, and therefore easy to guarantee the same result of  $M_\tau$  having rank 3.

*Remark 4.* If  $M_\tau$  has rank 3, it is straightforward to verify that up to two simultaneous thruster faults can be identified. However, in the case of three simultaneous thruster faults, it is, in principle, impossible to distinguish this fault from the one where the remaining single thruster occurs fault.

*Remark 5.* It is also worth noting that although no component of  $\tau_c$  is exactly zero, it still possible for one of  $\tau_i$ ,  $i = 1, \dots, 4$ , to be equal or near zero. In this case, from (24), the corresponding column vector in  $M_\tau$  becomes one with components of zero or very small magnitude. Consequently, the thrust loss associated with this column vector may not be sufficiently reflected in the residual, making it undetectable. To the best of the authors' knowledge, such detection failure is expected to be challenging regardless of any additional estimation methods applied.

If  $\tau$  is known after  $t > t_c$ , then through (24), it's possible for us to identify  $\delta W$ . The problem is that, in practice, it's impossible to know  $\tau$  since  $W$  is unknown after  $t > t_c$ . To solve this problem, we carry out a simulation study using BlueROV2 model [22] and investigate the similar case. In the simulation, the control law (20) is applied with the adaptation law taken as (21). Other detailed simulation conditions will be illustrated in Section IV.

Simulation results are shown in Fig. 2–4. Fig. 2 shows the sequential occurring of thruster faults, and corresponding residual changing trend is presented in Fig. 3, from which we can see that the constructed residual is suitable to detect the thruster faults. An additional point to note in Fig. 3 is that, after sparking at the time of  $\Delta W$  occurrence,  $R_{detection}$  automatically decreases again. This is a significantly different trend from the case in the authors' previous work [19], where

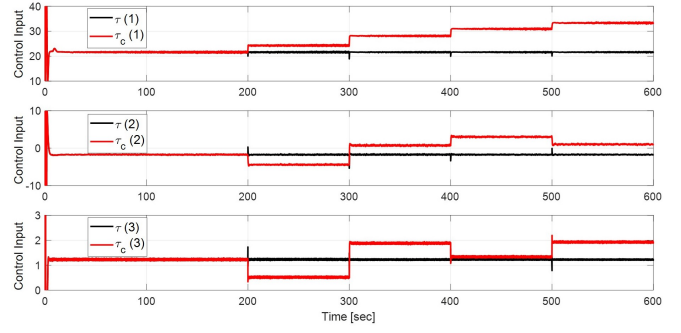


Fig. 4. Corresponding time variations of designed controller  $\tau_c$  and actual control input  $\tau$ .

the residual stays elevated if  $\hat{W}$  remains unchanged. To the authors' knowledge, this kind of auto-decreasing of residual is due to the adaptation component  $\hat{d}$  in the control law (20). Compared to the previous work [19], this kind of auto-decreasing of residual imposes two challenges; one is making it difficult to evaluate the magnitude of  $\Delta W$ , and the other is that the criterion for adjusting  $\hat{W}$  disappears. So we need to explore a new criterion for fault identification that can replace the residual in [19]. Fortunately, from Fig. 4, we can see that: before thruster faults occurrence ( $t < 200s$ ),  $\tau_c$  is exactly the same as  $\tau$ , and along with the sequential occurrences of thruster faults, compared to  $\tau_c$ , which correspondingly changes over the faults,  $\tau$  always maintains its original value as in  $t < 200s$  (except for the sparking at the very moment of fault occurrences). This also coincides with the result in Fig. 3. To conclude, the actual control input  $\tau$  after thruster faults occurrence can be approximated as  $\tau_c$  in the steady state case before thruster fault. Therefore, (24) can be rewritten as

$$M_\tau \delta W = \tau_c|_{t>t_c} - \tau_c|_{t<t_c} + w, \quad (26)$$

where  $w = \tau - \tau_c|_{t>t_c}$ .

In this paper,  $w$  is assumed to be a zero mean random noise. Since  $M_\tau$  and  $\tau_c$  are always known, now we can identify the thruster faults  $\delta W$ .

1) *Fault Detection:* Residual  $R_{detection}$  is defined as follows [19]:

$$R_{detection} = \sqrt{x_e^2 + y_e^2 + c_1 \psi_e^2}, \quad (27)$$

where  $c_1 > 0$  is a normalization constant parameter. At each time  $t > 0$ , fault detection for the calculated residual  $R_{detection}$  is simple and can be described as follows:

```

if ( $R_{detection} > c_2$ )&&(bFirst == 1)
    faultTrig = true;
    bFirst = 0;
end if

```

where  $c_2 > 0$  is a constant criterion.

2) *Fault Identification:* Fault identification follows the detection phase with  $faultTrig = true$ , see Algorithm 1.

In Algorithm 1,  $\tau_c$  represents both of  $\tau_c|_{t<t_c}$  and  $\tau_c|_{t>t_c}$ ;  $c_3, c_4 > 0$  are criterion parameters;  $T_3 > T_2 > T_1 > t_c$  are time-related parameters, where  $t_c$  represents the fault

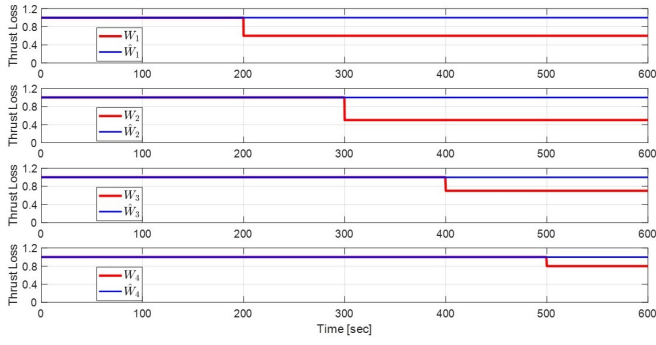


图2. 推进器故障发生场景。

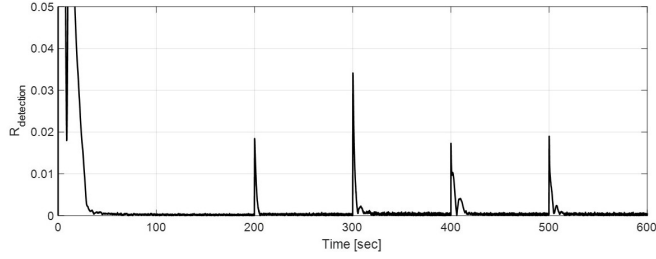


图3. 对应的残差测量。

$\tau_i$  非零, 因此容易保证  $M_\tau$  的秩为 3 的相同结果。

*Remark 4.* 如果  $M_\tau$  的秩为 3, 那么可以直接验证最多可以识别两个同时发生的推进器故障。然而, 在三个推进器同时发生故障的情况下, 原则上无法将此故障与剩余单个推进器发生故障的情况区分开来。

*Remark 5* 同样值得注意的是, 尽管  $\tau_c$  的任何组成部分都不完全为零, 但  $\tau_i, i=1, \dots, 4$  中的某一项仍可能等于或接近零。在这种情况下, 根据(24),  $M_\tau$  中相应的列向量将变为其分量为零或极小的向量。因此, 与该列向量相关的推力损失可能在残差中未得到充分反映, 从而难以检测。据作者所知, 无论采用任何额外的估计方法, 这种检测失败都可能是具有挑战性的。

如果在  $t > t_c$  之后已知  $\tau$ , 那么通过 (24), 我们就有可能确定  $\delta W$ 。问题是, 在实际中, 由于在  $t > t_c$  之后  $W$  是未知的, 所以不可能知道  $\tau$ 。为了解决这个问题, 我们使用 BlueROV2 模型 [22] 进行了仿真研究, 并调查了类似的情况。在仿真中, 控制律 (20) 被应用, 同时适应律采用 (21)。其他详细的仿真条件将在第四节中说明。

仿真结果如图2-4所示。图2展示了推进器故障的顺序发生, 图3给出了相应的残差变化趋势, 从中我们可以看出所构建的残差适合检测推进器故障。图3中需要注意的另一个点是, 在  $\Delta W$  发生时火花放电之后,  $R_{\text{detection}}$  会自动再次下降。这一趋势与作者之前工作[19]中的情况明显不同, 其中

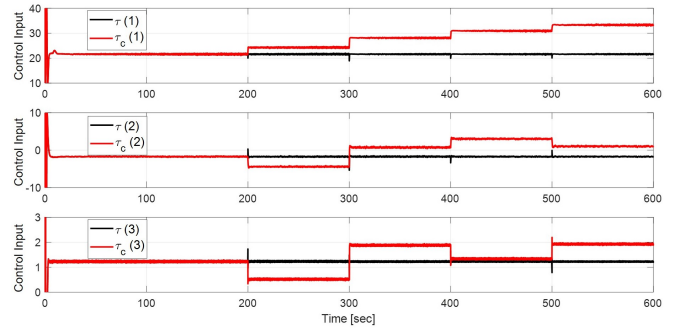


图4. 设计控制器  $\tau_c$  与实际控制输入  $\tau$  的对应时间变化。

如果  $\hat{W}$  保持不变, 残差仍会升高。根据作者的了解, 这种残差的自动下降是由于控制律 (20) 中的适应组件  $\hat{d}$  引起的。与之前的工作 [19] 相比, 这种残差的自动下降带来了两个挑战; 一是使得评估  $\Delta W$  的大小变得困难, 二是调整  $\hat{W}$  的标准消失。因此, 我们需要探索一种新的故障识别标准, 以替代 [19] 中的残差。幸运的是, 从图 4 可以看出: 在推进器故障发生之前 ( $t < 200s$ ),  $\tau_c$  与  $\tau$  完全相同, 随着推进器故障的连续发生, 相比于  $\tau_c$ , 其随故障相应变化,  $\tau$  始终保持其原始值, 如  $t < 200s$  (所示, 除了在故障发生的那一刻) 的火花。这也与图 3 的结果一致。总之, 实际控制输入

$$M_\tau \delta W = \tau_c|_{t>t_c} - \tau_c|_{t<t_c} + w, \quad (26)$$

在哪里  $w = \tau - \tau_c|_{t>t_c}$ 。

在本文中, 假设  $w$  为零均值随机噪声。由于  $M_\tau$  和  $\tau_c$  总是已知的, 现在我们可以识别推动器故障  $\delta W$ 。

1) *Fault Detection:* 残差  $R_{\text{detection}}$  定义如下 [19]:

$$R_{\text{detection}} = \sqrt{x_e^2 + y_e^2 + c_1 \psi_e^2}, \quad (27)$$

其中  $c_1 > 0$  是一个归一化常数参数。在每个时间  $t > 0$ , 对于计算得到的残差  $R_{\text{detection}}$  的故障检测很简单, 可以描述如下:

```

if ( $R_{\text{detection}} > c_2$ ) && (bFirst == 1)
    faultTrig = true;
    bFirst = 0;
end if

```

其中  $c_2 > 0$  是一个常数标准。

2) *Fault Identification:* 故障识别紧随  $\text{faultTrig} = \text{true}$  的检测阶段之后, 见算法1。

在算法 1 中,  $\tau_c$  表示  $\tau_c|_{t<t_c}$  和  $\tau_c|_{t>t_c}$ ;  $c_3, c_4 > 0$  是判定参数;  $T_3 > T_2 > T_1 > t_c$  是与时间相关的参数, 其中  $t_c$  表示故障

**Algorithm 1: Fault Identification ( )**


---

**Input:**  $M_\tau, \tau_c, \hat{W}$

```

1 waiting for faultTrig=true
2  $dW_1 = 0; dW_2 = [0; 0]; nC = 0;$ 
3 while (faultTrig)
4   if ( $t - t_c > T_1$ )
5      $[\delta W_n, rmse, n] = tFaultID(M_\tau, \tau_c, \hat{W}, single)$ 
6     if  $rmse \leq c_3$ 
7        $dW_{1+} = \delta W_n;$ 
8        $nC++;$ 
9       if  $t - t_c == T_2$ 
10         $\hat{W}(n, n) = \hat{W}(n, n)(1 - dW_1/nC);$ 
11      end if
12    else
13       $[\delta W_{n_1}, \delta W_{n_2}, rmse, n_1, n_2] =$ 
14         $tFaultID(M_\tau, \tau_c, \hat{W}, double)$ 
15      if  $rmse \leq c_4$ 
16         $dW_{2+} = [\delta W_{n_1}; \delta W_{n_2}];$ 
17         $nC++;$ 
18        if  $t - t_c == T_2$ 
19           $\hat{W}(n_1, n_1) = \hat{W}(n_1, n_1)(1 - dW_2[1]/nC);$ 
20           $\hat{W}(n_2, n_2) = \hat{W}(n_2, n_2)(1 - dW_2[2]/nC);$ 
21        end if
22      end if
23    end if
24    if  $t - t_c == T_3$ 
25      faultTrig=false;
26      bFirst=1;
27    end if
28  end while
29 end waiting for
30 return

```

---

occurrence time. At each  $t = t_c$ , the designed control input  $\tau_c$  experiences magnitude jumps, and  $T_1$  is used to allow for the convergence of this jumped  $\tau_c(t)$ . Due to the presence of random noise  $w$  as in (26), the estimated thrust loss  $\delta W_i$  also contains random noise. To mitigate this effect,  $T_2$  is used to attenuate the impact of this noise by simply averaging the estimation  $\delta W_i$  over the period  $T_2 - T_1$ . Each occurrence of control allocation at  $t = t_c + T_2$  using updated thrust loss parameter  $\hat{W}$  is typically followed by a spike in  $R_{detection}$ . To account for this,  $T_3$  is applied to allow for the re-convergence of  $R_{detection}$  after  $t = t_c + T_2$ .

To illustrate the core function  $tFaultID()$  in Algorithm 1, we rewrite (26) as follows:

$$\begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 \\ -\tau_1 & \tau_2 & \tau_3 & -\tau_4 \\ -a\tau_1 & a\tau_2 & -a\tau_3 & a\tau_4 \end{bmatrix} \begin{bmatrix} \delta W_1 \\ \vdots \\ \delta W_4 \end{bmatrix} = b_\tau, \quad (28)$$

where  $b_\tau = \tau_c|_{t>t_c} - \tau_c|_{t<t_c}$ .

From (28), we can see that in the case of single thruster fault, identification problem becomes to find out a vector among four column vectors of  $M_\tau$ , which is parallel to  $b_\tau$  and points in the same direction as  $b_\tau$ . Corresponding pseudo code for this process is shown in Algorithm 2, where  $rmse$

means root mean square error.

**Algorithm 2:  $tFaultID(M_\tau, \tau_c, \hat{W}, single)$** 


---

```

1 for  $i=1:4$ 
2    $\delta W[i] = M_\tau^\dagger(:, i)b_\tau;$ 
3    $rmse[i] = \sqrt{\frac{1}{3} \sum_{j=1}^3 [b_\tau(j) - \delta W[i]M_\tau(j, i)]^2};$ 
4 end for
5 return minimum  $rmse[i]$  and corresponding  $\delta W[i]$  and  $i$ 

```

---

In the case of double thruster faults, the identification process is similar to that in Algorithm 2, except that the column vector in Algorithm 2 is replaced by a combination of two column vectors from  $M_\tau$ . Given four column vectors  $M_\tau(:, i)$ ,  $i = 1, \dots, 4$ , there are a total of six different combinations of two columns. Let  $\Omega$  denotes a  $3 \times 2$  matrix set consisting of these six different combinations. The identification process for double thruster faults can then be described as in Algorithm 3.

**Algorithm 3:  $tFaultID(M_\tau, \tau_c, \hat{W}, double)$** 


---

```

1 Construct  $\Omega$  from  $M_\tau$ ;
2 for  $i=1:6$ 
3    $a_i \in \Omega;$ 
4    $\delta W(:, i) = a_i^\dagger(:, i)b_\tau;$ 
5    $rmse[i] = \sqrt{\frac{1}{3} \sum_{j=1}^3 [b_\tau(j) - \delta W(1, i)a_i(j, 1) - \delta W(2, i)a_i(j, 2)]^2};$ 
6 end for
7 return minimum  $rmse[i]$  and corresponding  $\delta W(:, i)$  and  $i$ 

```

---

In Algorithm 3, from returned case number  $i$ , we can further decode the corresponding column numbers  $j, k \in \{1, \dots, 4\}$  with  $a_i = [M_\tau(:, j); M_\tau(:, k)]$ .

3) *Control Allocation*: Control allocation can be efficiently achieved using (10), where  $\hat{W}$  is replaced with the estimated values as specified in Algorithm 1 (line 10 or line 18 and 19).

## IV. NUMERICAL STUDIES

The 3DOF horizontal dynamic positioning of an underwater vehicle in a strong sea current environment is considered in this numerical studies. As mentioned before, the 6DOF nonlinear model of BlueROV2 Heavy ROV [22] is directly applied in the Matlab-based simulation. The sea current is simulated with  $u_c = 1.0m/s$  and  $\psi_c = -120^\circ$ , while the remaining uncertainty terms are set as  $d - d_{sc} = [1 - 2rand; 1 - 2rand; 0.03(1 - rand)]$  with  $d_{sc}$  indicating the aforementioned sea current. Here,  $rand \in [0, 1]$  denotes a random variable.

For the vehicle's thruster configuration, we have  $K = diag(40, 40, 40, 40)$ ,  $\alpha = 45^\circ$  and  $l = 0.1888m$ . The vehicle's dynamic positioning reference configuration is set as  $\eta_d = [10m, 2m, 70^\circ]^T$ , and the vehicle's initial state is chosen as  $X(0) = \mathbf{0}^{12 \times 1}$  (in the simulation, 6DOF model is used instead of 3DOF). Controller design parameters are set as  $\Gamma_1 = diag(1, 1, 1500)$ ,  $\Gamma_2 = diag(1, 1, 0.6)$ ,

**Algorithm 1: Fault Identification ( )**


---

**Input:**  $M_\tau, \tau_c, \hat{W}$

```

1 waiting for faultTrig=true
2  $dW_1 = 0; dW_2 = [0; 0]; nC = 0;$ 
3 while (faultTrig)
4   if ( $t - t_c > T_1$ )
5      $[\delta W_n, rmse, n] = tFaultID(M_\tau, \tau_c, \hat{W}, single)$ 
6     if  $rmse \leq c_3$ 
7        $dW_{1+} = \delta W_n;$ 
8        $nC++;$ 
9       if  $t - t_c == T_2$ 
10         $\hat{W}(n, n) = \hat{W}(n, n)(1 - dW_1/nC);$ 
11      end if
12    else
13       $[\delta W_{n_1}, \delta W_{n_2}, rmse, n_1, n_2] =$ 
         $tFaultID(M_\tau, \tau_c, \hat{W}, double)$ 
14      if  $rmse \leq c_4$ 
15         $dW_{2+} = [\delta W_{n_1}; \delta W_{n_2}];$ 
16         $nC++;$ 
17        if  $t - t_c == T_2$ 
18           $\hat{W}(n_1, n_1) = \hat{W}(n_1, n_1)(1 - dW_2[1]/nC);$ 
19           $\hat{W}(n_2, n_2) = \hat{W}(n_2, n_2)(1 - dW_2[2]/nC);$ 
20        end if
21      end if
22    end if
23  end if
24  if  $t - t_c == T_3$ 
25    faultTrig=false;
26    bFirst=1;
27  end if
28 end while
29 end waiting for
30 return

```

---

发生时间。在每个  $t = t_c$ ，设计的控制输入  $\tau_c$  会出现幅值跳变，并且使用  $T_1$  来允许这种跳变的  $\tau_c(t)$  收敛。由于存在如 (26) 所示的随机噪声  $w$ ，估计的推力损失  $\delta W_i$  也包含随机噪声。为了减轻这种影响，使用  $T_2$  通过简单地在周期  $T_2 - T_1$  内平均估计  $\delta W_i$  来衰减这种噪声的影响。每次在  $t = t_c + T_2$  使用更新的推力损失参数  $\hat{W}$  进行控制分配时，通常会随后出现  $R_{detection}$  的尖峰。为了解决这一问题，应用  $T_3$  以允许  $R_{detection}$  在  $t = t_c + T_2$  后重新收敛。

为了说明算法1中核心函数  $tFaultID()$ ，我们将(26)重写如下：

$$\begin{bmatrix} \tau_1 & \tau_2 & \tau_3 & \tau_4 \\ -\tau_1 & \tau_2 & \tau_3 & -\tau_4 \\ -a\tau_1 & a\tau_2 & -a\tau_3 & a\tau_4 \end{bmatrix} \begin{bmatrix} \delta W_1 \\ \vdots \\ \delta W_4 \end{bmatrix} = b_\tau, \quad (28)$$

在哪里  $b_\tau = \tau_c|_{t>t_c} - \tau_c|_{t<t_c}$

从 (28) 可以看出，在单推进器故障的情况下，识别问题变为在  $M_\tau$  的四个列向量中找到一个向量，该向量与  $b_\tau$  平行，并且与  $b_\tau$  方向相同。该过程的相应伪代码如算法 2 所示，其中  $rmse$

表示均方根误差。

**Algorithm 2:  $tFaultID(M_\tau, \tau_c, \hat{W}, single)$** 


---

```

1 for  $i=1:4$ 
2    $\delta W[i] = M_\tau^\dagger(:, i)b_\tau;$ 
3    $rmse[i] = \sqrt{\frac{1}{3} \sum_{j=1}^3 [b_\tau(j) - \delta W[i]M_\tau(j, i)]^2};$ 
4 end for
5 return minimum  $rmse[i]$  and corresponding  $\delta W[i]$  and  $i$ 

```

---

在双推进器故障的情况下，识别过程类似于算法2，只是算法2中的列向量被来自  $M_\tau$  的两个列向量的组合所取代。给定四个列向量  $M_\tau(:, i)$ ,  $i = 1, \dots, 4$ ，总共有六种不同的两列组合。让  $\Omega$  表示一个由这六种不同组合组成的  $3 \times 2$  矩阵集合。双推进器故障的识别过程可以按照算法3进行描述。

**Algorithm 3:  $tFaultID(M_\tau, \tau_c, \hat{W}, double)$** 


---

```

1 Construct  $\Omega$  from  $M_\tau;$ 
2 for  $i=1:6$ 
3    $a_i \in \Omega;$ 
4    $\delta W(:, i) = a_i^\dagger(:, i)b_\tau;$ 
5    $rmse[i] = \sqrt{\frac{1}{3} \sum_{j=1}^3 [b_\tau(j) - \delta W(1, i)a_i(j, 1) - \delta W(2, i)a_i(j, 2)]^2};$ 
6 end for
7 return minimum  $rmse[i]$  and corresponding  $\delta W(:, i)$  and  $i$ 

```

---

在算法3中，从返回的案例编号  $i$ ，我们可以进一步解码相应的列号  $j, k \in \{1, \dots, 4\}$  与  $a_i = [M_\tau(:, j); M_\tau(:, k)]$ 。

3) *Control Allocation*: 可以使用 (10) 高效地实现控制分配，其中  $\hat{W}$  被替换为算法 1 中指定的估计值（第 10 行或第 18 和第 19 行）。

## IV. 数值研究

在本数值研究中，考虑了水下车辆在强海流环境下的3自由度水平动态定位。如前所述，直接在基于Matlab的仿真中应用了BlueROV2 Heavy ROV [22]的6自由度非线性模型。海流模拟为  $u_c = 1.0m/s$  和  $\psi_c = -120^\circ$ ，而其余不确定性项设为  $d - d_{sc} = [1 - 2rand; 1 - 2rand; 0.03(1 - rand)]$ ，其中  $d_{sc}$  表示前文提到的海流。这里， $rand \in [0, 1]$  表示一个随机变量。

对于车辆的推进器配置，我们有  $K = diag(40, 40, 40, 40)$ ,  $\alpha = 45^\circ$  和  $l = 0.1888m$ 。车辆的动态定位参考配置设置为  $\eta_d = [10m, 2m, 70^\circ]^T$ ，车辆的初始状态选择为  $X(0) = \mathbf{0}^{12 \times 1}$ （在仿真中，使用6自由度模型而非3自由度）。控制器设计参数设定为  $\Gamma_1 = diag(1, 1, 1500)$ ,  $\Gamma_2 = diag(1, 1, 0.6)$ ,

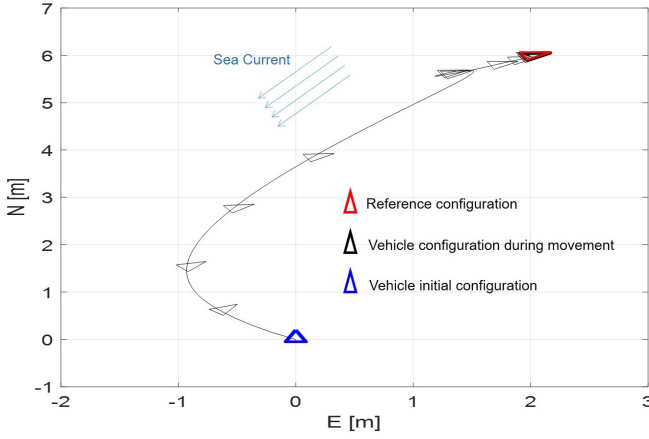


Fig. 5. Vehicle trajectory during the dynamic positioning without thrust loss parameter update.

$A_1 = \text{diag}(1.5, 1.5, 5)$ ,  $A_2 = \text{diag}(1, 1, 12)$ ,  $\hat{D}_m = \mathbf{0}^{3 \times 1}$ , and  $\epsilon_1 = \epsilon_2 = \epsilon_3 = 0.1$ . For the smooth function  $\varphi(\cdot)$ , it is chosen as  $\varphi(\xi) = \tanh(\kappa\xi/\epsilon)$  in the simulation with  $\kappa = 0.2785$  [25]. FDI algorithm related parameters are chosen as:  $c_1 = (180/\pi)^2$ ,  $c_2 = 0.005$ ,  $c_3 = c_4 = 0.1$ ,  $T_1 = 20s$ ,  $T_2 = 25s$ , and  $T_3 = 35s$ . Finally, the algorithm sampling time is set as  $0.01s$ .

In the simulation, we carry out various case studies, among which Fig. 2–4 present a scenario where sequential thruster faults occur as  $W_1 \rightarrow W_4$ , without control allocation based on thrust loss parameters estimation (line 10 in Algorithm 1 is not executed). Fig. 5 shows the vehicle's trajectory for this case study, which closely resembles the trajectory observed when no thruster fault occurred. Some interesting results from other case studies are summarized below.

1) *A Single Thruster Fault with Sequential Thrust Loss Leading to Failure*: The simulation results for this case are presented in Fig. 6 and 7. Fig. 6 demonstrates that the proposed FDI algorithm effectively detects and identifies faults while maintaining accurate estimation capability. Notably, as shown in Fig. 7, following the single thruster failure at  $t_c = 1000s$ , the residual exhibits a slight increase compared to the previous sequential stages, highlighted by the red-dotted circle. This suggests a discontinuity in control allocation at the moment when the thruster configuration transitions from four thrusters to three.

2) *Combination of Single and Double Thruster Faults*: The corresponding case study results are shown in Fig. 8, demonstrating that the proposed scheme effectively detects and identifies both single and double thruster faults while maintaining accurate estimation performance. In fact, we carried out a large number of similar case studies and observed consistent results throughout.

3) *Fault Detection Failure*: If we set  $\eta_d = [10, 2, 90^\circ]^T$  instead of  $\eta_d = [10, 2, 70^\circ]^T$ , then  $M_\tau$  takes the following form:

$$M_\tau = \begin{bmatrix} 3.5178 & 5.0833 & -0.7511 & 9.3522 \\ -3.5178 & 5.0833 & -0.7511 & -9.3522 \\ -0.9393 & 1.3572 & 0.2006 & 2.4971 \end{bmatrix}, \quad (29)$$

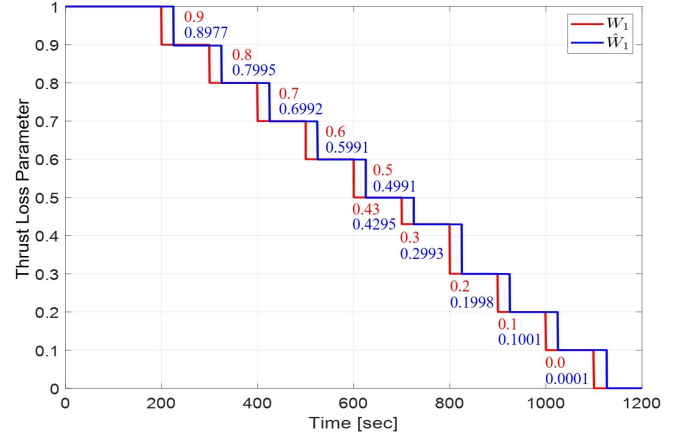


Fig. 6. Single thruster sequential faults and their detection, identification, and corresponding thrust loss estimations.

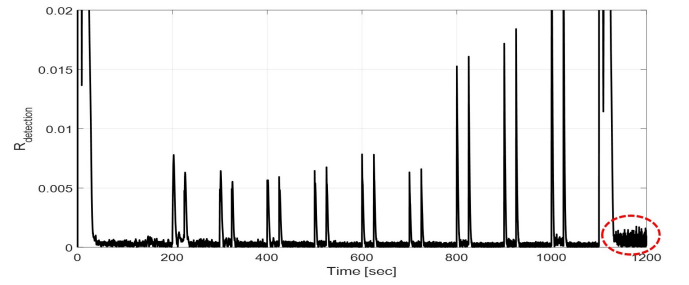


Fig. 7. Residual variation trend during sequential single thruster faults leading to failure.

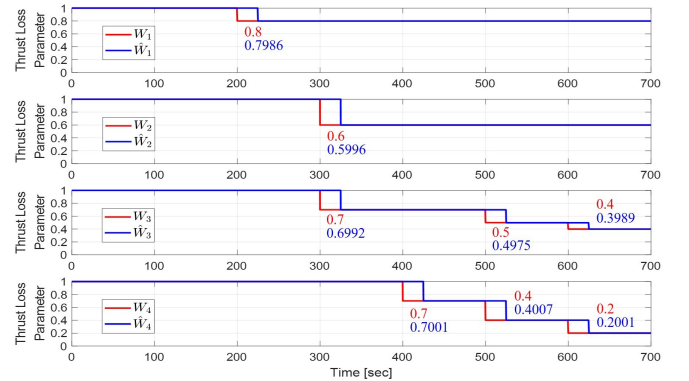


Fig. 8. Detection, identification, and thrust loss estimation for single and double thruster fault combinations.

where the third column has relatively small component magnitude compared to the others. In this case, if we set  $\delta W_3 = 0.1$ , indicating a relatively small thrust loss for the third thruster, Fig. 9 shows that the proposed FDI algorithm fails to detect this fault, which further causes the detection failure of  $W_4$  at  $t = 500s$ . However, when the thrust loss is increased to  $\delta W_3 = 0.3$ , both  $\delta W_3$  and  $\delta W_4$  becomes detectable, see Fig. 10. As mentioned in Remark 5, in the specific case of  $\tau_c$  which causes one of  $\tau_i$ ,  $i = 1, \dots, 4$  to be equal or near zero, the thrust loss in this  $i$ th thruster becomes difficult to be detected. This could be recognized as an inherent limitation of the FDI algorithm proposed in

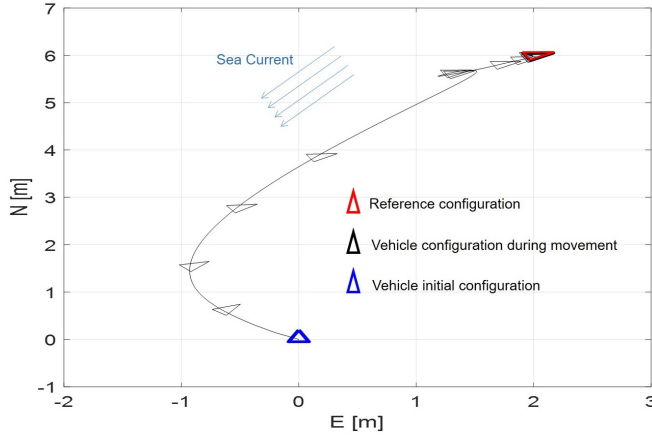


图5. 在未更新推力损失参数情况下的动态定位车辆轨迹。

$A_1 = \text{diag}(1.5, 1.5, 5)$ ,  $A_2 = \text{diag}(1, 1, 12)$ ,  $\hat{D}_m = \mathbf{0}^{3 \times 1}$ , 和  $\epsilon_1 = \epsilon_2 = \epsilon_3 = 0.1$ 。对于光滑函数  $\varphi(\cdot)$ , 在仿真中选择为  $\varphi(\xi) = \tanh(\kappa\xi/\epsilon)$ , 并且  $\kappa = 0.2785$  [25]。FDI 算法相关参数选择为:  $c_1 = (180/\pi)^2$ ,  $c_2 = 0.005$ ,  $c_3 = c_4 = 0.1$ ,  $T_1 = 20s$ ,  $T_2 = 25s$ , 和  $T_3 = 35s$ 。最后, 算法采样时间设置为  $0.01s$ 。

在仿真中, 我们进行了各种案例研究, 其中图 2-4 展示了一个情况下连续推进器故障发生为  $W_1 \rightarrow W_4$  的场景, 并且没有基于推力损失参数估计的控制分配 (算法 1 中的第 10 行未执行)。图 5 显示了该案例研究的车辆轨迹, 其与未发生推进器故障时观察到的轨迹非常相似。其他案例研究的一些有趣结果总结如下。

1) *A Single Thruster Fault with Sequential Thrust Loss Leading to Failure*: 本案例的仿真结果如图6和图7所示。图6表明所提出的FDI算法能够有效地检测和识别故障, 同时保持准确的估计能力。值得注意的是, 如图7所示, 在  $t_c = 1000s$  单个推进器故障发生后, 相较于之前的连续阶段, 残差略有增加, 红色虚线圆圈标出的位置。这表明在推进器配置从四个推进器过渡到三个推进器的时刻, 控制分配存在不连续性。

2) *Combination of Single and Double Thruster Faults*: 相应的案例研究结果如图8所示, 表明所提出的方案能够有效地检测和识别单个和双推进器故障, 同时保持准确的估计性能。实际上, 我们进行了大量类似的案例研究, 并在整个过程中观察到一致的结果。

3) *Fault Detection Failure*: 如果我们设置  $\eta_d = [10, 2, 90^\circ]^T$  而不是  $\eta_d = [10, 2, 70^\circ]^T$ , 那么  $M_\tau$  的形式如下:

$$M_\tau = \begin{bmatrix} 3.5178 & 5.0833 & -0.7511 & 9.3522 \\ -3.5178 & 5.0833 & -0.7511 & -9.3522 \\ -0.9393 & 1.3572 & 0.2006 & 2.4971 \end{bmatrix}, \quad (29)$$

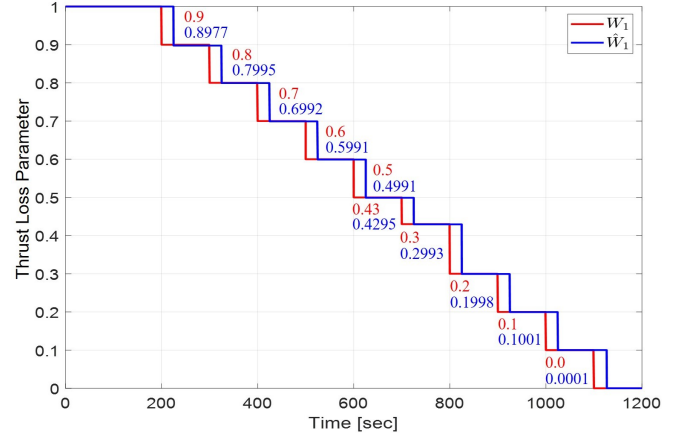


图6. 单个推进器顺序故障及其检测、识别和相应的推力损失估计。

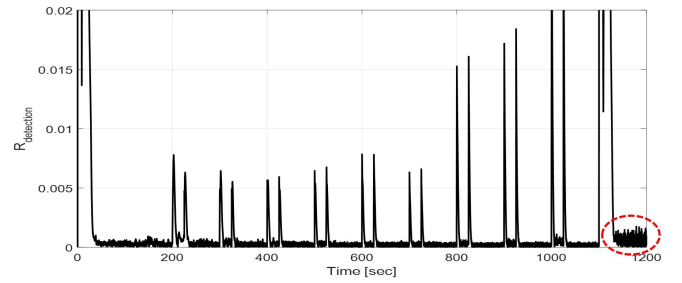


图7. 导致故障的连续单推进器故障期间残余变化趋势。

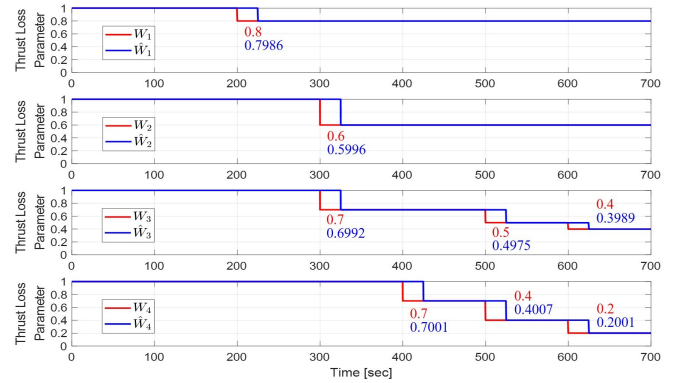


图8. 单推进器和双推进器故障组合的检测、识别及推力损失估计。

第三列的分量幅值相较于其他列较小。在这种情况下, 如果我们设定  $\delta W_3 = 0.1$ , 表示第三个推进器的推力损失相对较小, 图 9 显示所提出的 FDI 算法未能检测到该故障, 这进一步导致在  $t = 500s$  时  $W_4$  的检测失败。然而, 当推力损失增加到  $\delta W_3 = 0.3$  时,  $\delta W_3$  和  $\delta W_4$  都能够被检测到, 见图 10。如在备注 5 中提到的, 在  $\tau_c$  的特定情况下, 这会导致  $\tau_i$  中的  $i = 1, \dots, 4$  等于或接近零, 该推进器的推力损失变得难以检测。这可以被认为是所提出 FDI 算法的固有限制。

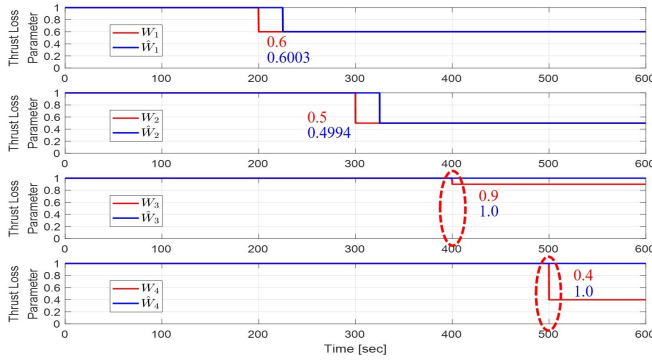


Fig. 9. Failure of fault detection in the case of a small thrust loss in the thruster, which corresponds to the third column vector of  $M_T$  in (29), having small magnitude components.

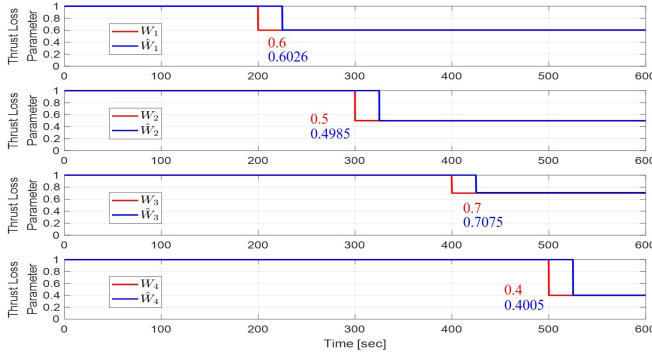


Fig. 10. Successful fault detection with increasing thrust loss in the third thruster, while maintaining the same conditions as in Fig. 9.

this paper.

## V. CONCLUSION

This paper has presented a novel FTC scheme for a class of underwater vehicles with thruster redundancy. Rather than relying on estimation errors, the residual is constructed by directly applying MCE, eliminating the need for an additional estimation system. Through detailed and precise analyses of MCE variation trends in the presence of thruster faults, valuable information regarding the actual control input after a thruster faults can be acquired, allowing us to formulate both the fault identification method and the thrust loss estimations. The proposed FDI algorithm can handle up to two simultaneous thruster faults. Various numerical studies have been carried out, demonstrating that the proposed method provides satisfactory FDI performance and accurate thrust loss estimation.

Since the thruster configuration matrix is  $3 \times 4$ , in principle, simultaneous faults in three thrusters cannot be distinguished from a fault in the remaining single thruster. However, by restricting each thrust loss to the domain  $[0, 1]$ , it may be possible to solve this triple-fault problem. Therefore, this will be an interesting topic for our future research.

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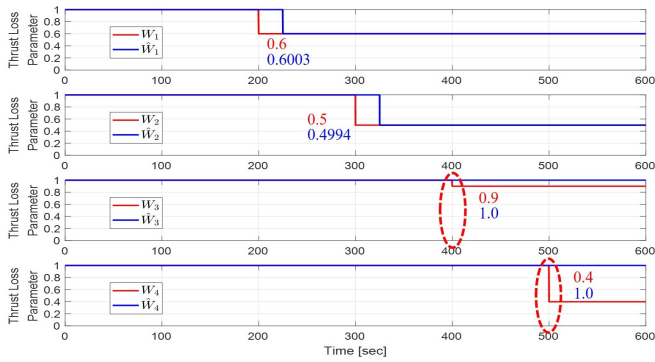


图9. 在推进器出现小推力损失的情况下故障检测失败，这对应于(29)中 $M_r$ 的第三列向量，其分量幅值较小。

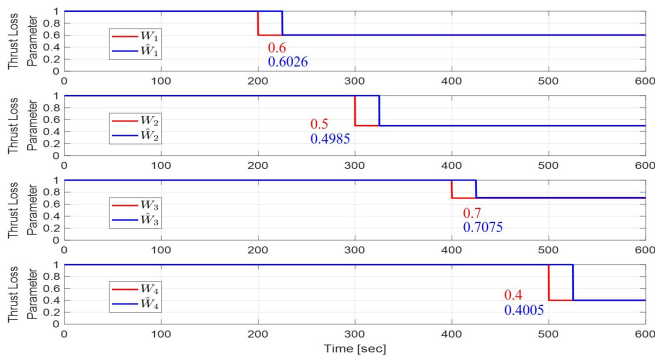


图10. 在第三个推进器推力损失增加的情况下成功检测故障，同时保持与图9相同的条件。

这篇论文。

## 五、结论

本文提出了一种针对具有推进器冗余水下车辆的创新型容错控制（FTC）方案。该方案不是依赖于估计误差，而是通过直接应用最大模值准则（MCE）构建残差，从而无需额外的估计系统。通过对推进器故障情况下MCE变化趋势的详细和精确分析，可以获取关于推进器故障后实际控制输入的有价值信息，从而制定故障识别方法和推力损失估计。所提出的故障检测与隔离（FDI）算法最多可处理同时发生的两个推进器故障。进行了多种数值研究，结果表明该方法在FDI性能和推力损失估计方面均表现令人满意。

由于推进器配置矩阵是 $3 \times 4$ ，原则上，三个推进器的同时故障无法与剩余单个推进器的故障区分开来。然而，通过将每个推力损失限制在 $[0, 1]$ 的范围内，可能有机会解决这个三重故障问题。因此，这将是未来研究的一个有趣课题。

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